

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EX-8004 (2) (CBGS)

B.E. VIII Semester Examination, June 2020

Choice Based Grading System (CBGS)

Power Electronics Converters for Renewable Energy

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What do you mean by conventional , nonconventional , renewable and nonrenewable energy? Explain in brief.
b) Write the significance of renewable sources and their exploitation in energy planning.
2. a) Draw layout of a standalone solar PV system.
b) What do you mean by energy efficiency? Discuss the status of world energy resources.
3. a) Explain the concept of PWM techniques single and multiples form.
b) Discuss the working of UPS.
4. a) What do you understand by DC-DC buck boost converter?
b) Discuss the bidirectional converter.
5. Explain single phase and three phase inverters with proper circuit diagram and their waveforms.
6. Explain three phase to single phase and single phase to three phase cyclo-converter with proper circuit diagram and waveforms.
7. Explain wind power system in details. Also plot and explain PV and IV characteristic for the wind energy.
8. Explain the following (any two)
 - i) Mitigation of harmonics.
 - ii) Active and Passive filters.
 - iii) Synchronous generator with back to back controlled converter.
 - iv) Doubly fed induction generator with rotor side converter topologies.
